

Table 1: Access Control List

Internal groups	External access requirements	Port Alias Group	External IP Alias Group
Internal network	DNS servers	TCP, UDP – 53	DNS servers
Admin	Mailing services to access Gmail using the mail client	GmailServices	Gmail servers
	Additional mail server using SMTP and POP services	Mail services	mail.company-mail.com
	Web browsing	Internet services	Any
	Admin console of additional mail server	8085	www.company-mail.com
Engg	Mailing services to access Gmail using the mail client		Gmail servers
	Customer site FTP access	FTP services	ftp.customersite.com
Accounts	Mailing services to access Gmail using the mail client	GmailServices	Gmail servers
	LBT Pune website	8080	www.website.com

Please discuss in detail with computer users and have the patience to create these lists. The more details you get, the fewer reconfiguration calls will be needed later.

Identify internal IP addresses, external IP addresses, external host names and services required for controlling access, etc. For example, for allowing Gmail access, we need to configure three groups:

- GmailServices Port alias group containing the following TCP services (ports) required for Gmail access:

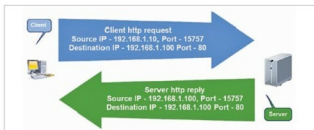


Figure 1: Concept of ports

- IMAPS – 993
 - SMTPTS – 465
 - POP3S – 993
 - Submission – 587
- GmailServers IP alias group containing IP addresses for the following servers:
 - smtp.gmail.com
 - imap.gmail.com
 - Group of internal systems required to access Gmail using the mail client.

Think about how to group various internal/external IP addresses and services to create the minimum number of access rules. Create IP and ports alias lists from *Firewall – Alias* menus.



Note: Planning before implementing rules is the best way – there are no shortcuts.

Once the aliases are ready, go to *Firewall – Rules – LAN* and proceed to create the desired access rules as per the requirements already defined in Table 1.

From the *Source* dropdown box, select *Single host or alias*. Type the name of the predefined alias in the box in front – pfSense will auto display all matching aliases.

Since this firewall is configured with dual WAN, click on *Display Advanced* under *Extra Options* and select *DualWAN Gateway*.

Similarly, create the additional required rules to allow traffic from the source towards the destination by using *Services*. For example, to allow the computer from Admin group to access Gmail using a mail client, create the following rule:

- Select the pass rule with source as Admin group.
- Select the destination with Gmail servers using Gmail services.
- Use the gateway and dual WAN to ensure this rule works with both the WAN interfaces.

Knowledgeable readers may revisit ACL in Table 1 before reading further, to check if any rule is missing.

At the time of installation, pfSense configures a default rule, which allows all traffic from the LAN net towards any destination. Once all rules are configured, disable this default rule by clicking the ✓ button.



Figure 2: GmailServices